

UCI Online Retail

Business Data Analysis — Executive Insights

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Executive Summary

The Bottom Line — Key Business Insights



Scale: 541,909 transactions analyzed (Dec 2010 – Dec 2011), yielding £10.64M in valid revenue across 4,338 known customers.



Seasonality: November 2011 generated £1.5M (2x monthly average) driven by a dual-driver phenomenon: core B2B restocking + Christmas-themed product surge.

Customer Concentration: Top 26% of customers generate 80% of revenue (Gini = 0.142) — a healthy, broad B2B wholesale base, not whale-dependent.



Strategic Risk: UK accounts for 84.6% of revenue AND 90% of Champions — a dual concentration crisis requiring international diversification.



Biggest Opportunity: 47.8% of customers are At Risk or Lost — targeted win-back campaigns could unlock ~£690K in annual revenue.

Data Integrity & Critical Decisions

Evidence-based choices that ensured analytical rigor

- ✓ Missing CustomerID (24.93%): Labeled 'Unknown' for sales tracking; strictly excluded from RFM/Pareto analysis (likely B2C guest checkouts).
- ✓ Exact Duplicates (5,268 rows): Removed to prevent artificial inflation of metrics.
- ✓ Extreme Outlier Isolated: Single invoice of 'PAPER CRAFT, LITTLE BIRDIE' (80,995 units, £168,469.60) — isolated, not deleted, preserving total revenue reconciliation.
- ✓ Operational Items Excluded: £362,268.47 in DOTCOM POSTAGE, POSTAGE, Manual, Bank Charges removed from product rankings (service fees, not merchandise).
- 📊 Final Clean Dataset: 522,728 physical product transactions representing £10,279,842.33 in true merchandise revenue.

Key Finding 1: Seasonal Revenue Drivers

November 2011 — A Dual-Driver Phenomenon

 November 2011 Revenue: £1,503,866.78 — approximately 2x the average monthly revenue.

 Driver A (Core Amplification): RABBIT NIGHT LIGHT generated £34,478.40 in November alone — 11.7x its monthly average (48.5% of annual revenue in one month).

 Driver B (Seasonal Emergence): Christmas-themed products (PAPER CHAIN KIT, CHILLI LIGHTS, HOT WATER BOTTLE KEEP CALM) contributed £50K+ in incremental revenue.

 Hybrid Model Confirmed: Only 40% overlap between November's Top 10 and annual Top 10 — predictable B2B restocking + distinct seasonal surge.

 Top November Product: RABBIT NIGHT LIGHT (£34,478.40, 487 invoices, Avg_Qty=29.65).



Key Finding 2: Customer Segmentation (RFM)

Pareto 26/80 | Gini 0.142 | 4 Behavioral Segments

 Pareto Validation: Top 26% of customers generate 80% of revenue — close to classic 80/20 rule.

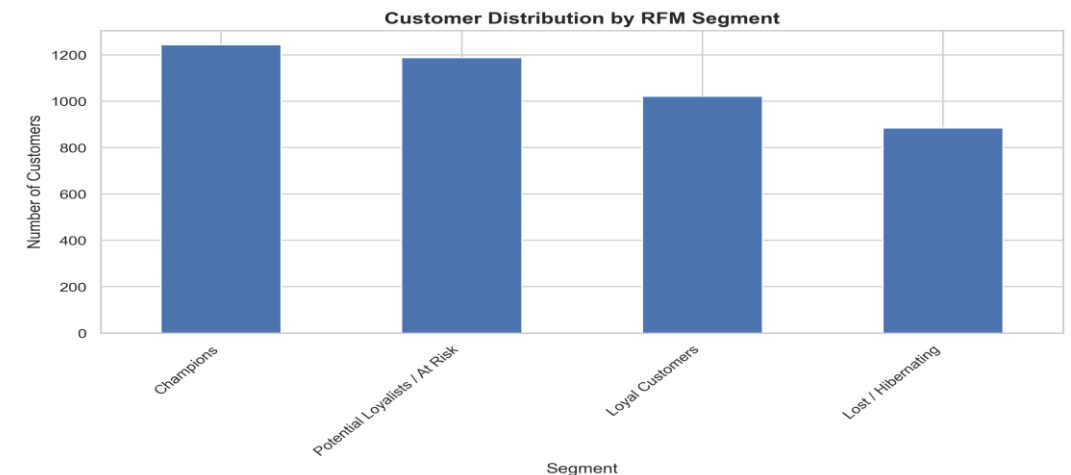
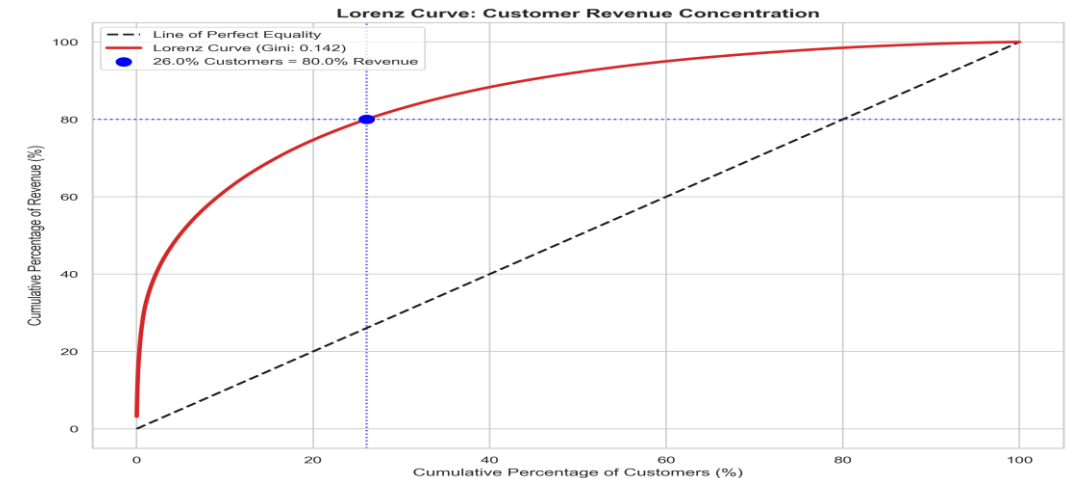
 Gini Coefficient = 0.142 (surprisingly low for B2B) — broad base of ~1,128 high-value wholesale customers, not whale-dependent.

 Champions (28.6% | 1,243): Avg £5,465 revenue, 10 orders/year, 18-day recency — the core B2B engine.

 At Risk (27.4% | 1,188): Avg £580 revenue, 102-day recency — massive retention opportunity.

 Lost/Hibernating (20.4% | 885): Avg £228 revenue, 223-day recency — minimal ROI on marketing spend.

 Critical Insight: 47.8% of customer base is At Risk or Lost — targeted win-back could add ~£690K annually.



Key Finding 3: Geographic Concentration Risk

Dual Concentration Crisis — UK Dependency

GB United Kingdom: 84.6% of total revenue (£9M) AND 90% of Champions (1,119 out of 1,243) — a dual concentration crisis.

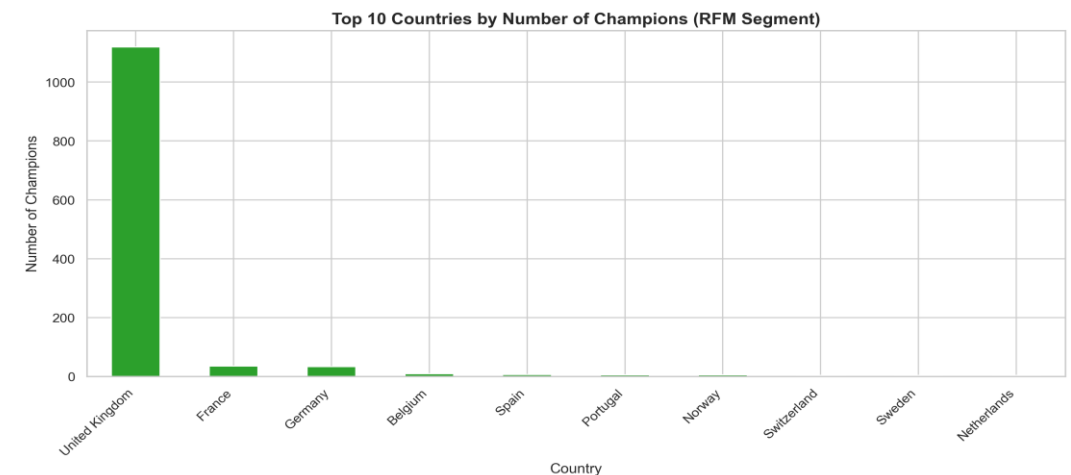
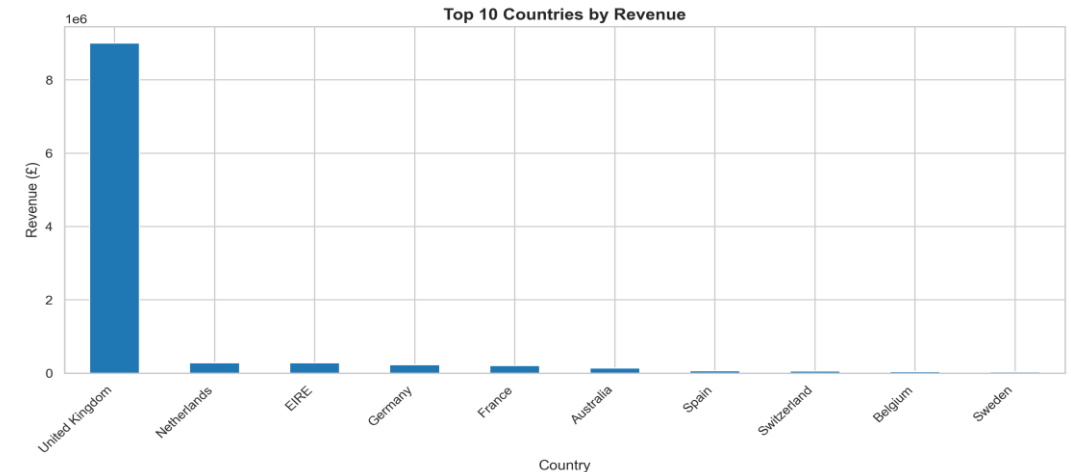
IE EIRE (Ireland): Only 4 customers generate £283K — average £70,785 per customer (highest per-customer value).

NL Netherlands: 9 customers generate £285K — average £31,716 per customer.

A Australia: Highest AOV (£2,429) with only 9 customers — premium niche market.

FR France & DE Germany: 35 and 33 Champions respectively — early traction for international expansion.

⚠ Risk: Any UK disruption (Brexit, recession) would simultaneously impact revenue volume AND customer quality.



Product Portfolio Analysis

B2B Wholesale Patterns & Dead Stock Alert

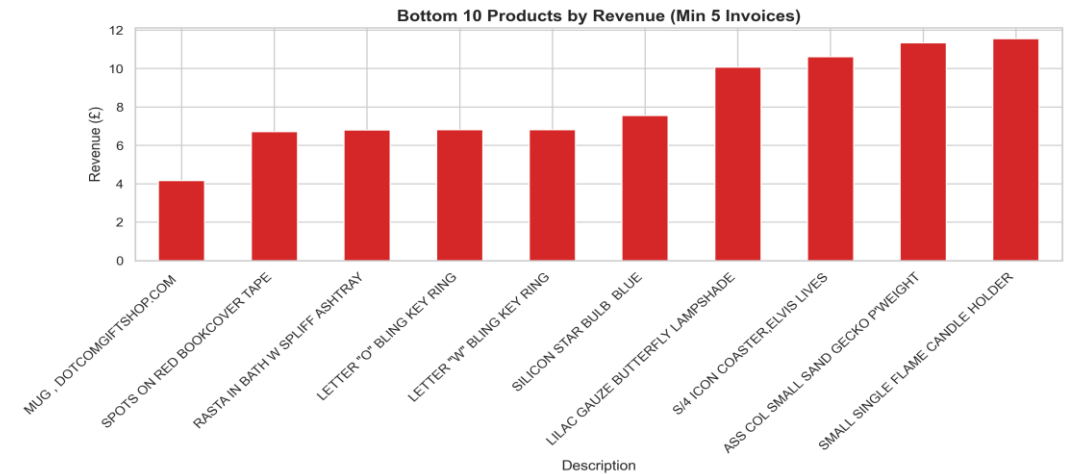
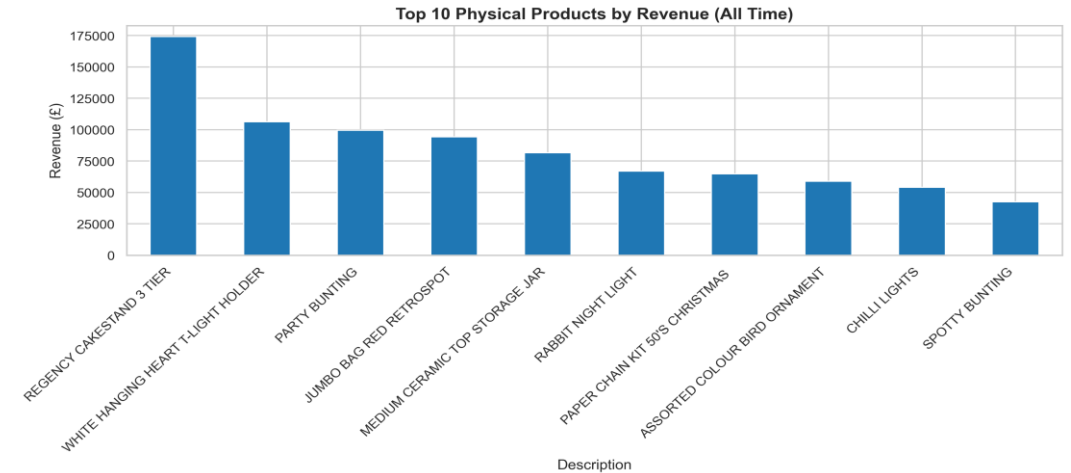
🏆 Top Revenue Product: REGENCY CAKESTAND 3 TIER (£174K across 1,988 invoices).

📦 Strong B2B Indicators (Avg_Qty > 20/invoice):

- MEDIUM CERAMIC TOP STORAGE JAR: 312 units/invoice
- RABBIT NIGHT LIGHT: 30 units/invoice
- JUMBO BAG RED RETROSPOT: 23 units/invoice

⚠️ Dead Stock Alert: Bottom 10 products generate only £4–£11 total — storage costs likely exceed revenue.

💡 Recommendation: Bundle or discontinue low-performers to free warehouse space.



Operational Revenue & Customer Behavior

Separating Service Fees from Product Performance



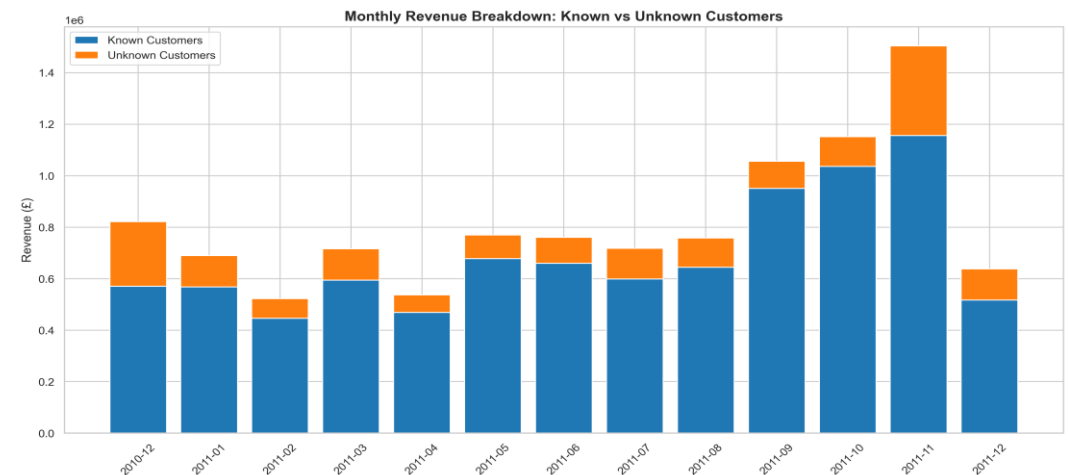
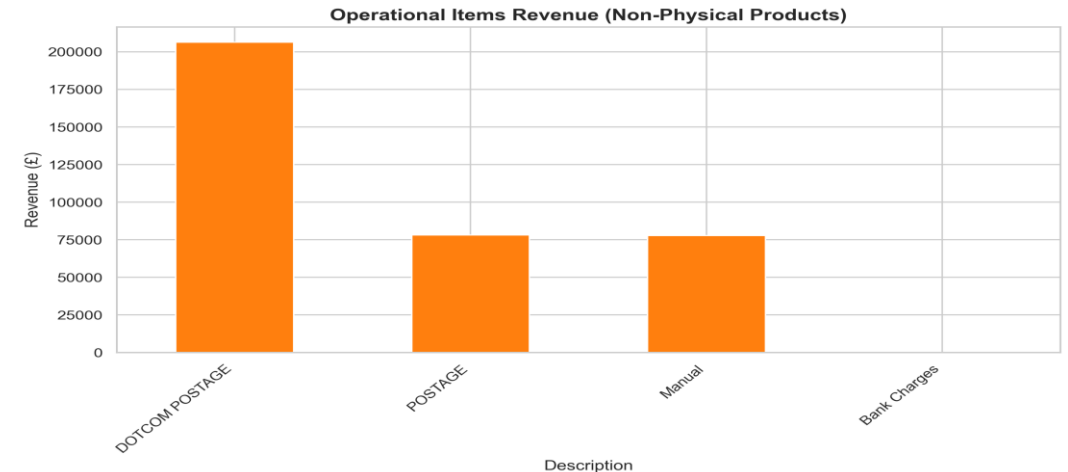
Operational Items Total: £362,268.47 (3.4% of revenue) — excluded from product rankings.

- DOTCOM POSTAGE: £206K (706 transactions, £292 avg) — per-order shipping fee.
- POSTAGE: £78K (3,150 units) — bulk shipping labels.
- Manual: £77K (6,985 units) — accounting adjustments, NOT products.



Unknown Customer Behavior: 16.5% of revenue from guest checkouts, spiking to 23.1% in November.

Insight: Seasonal spikes attract more guest buyers — opportunity for post-purchase conversion.



Strategic Recommendations

Prioritized Action Plan

HIGH PRIORITY (30 Days)

R1: VIP loyalty program for 1,119 UK Champions + dedicated account managers.

R2: Win-back email campaign for 1,188 At Risk customers (15% discount). Expected impact: ~£690K.

R3: Discontinue or bundle Bottom 10 products (<£12/year each) to free warehouse space.

MEDIUM PRIORITY (90 Days)

R4: Dual-track seasonal inventory — pre-build RABBIT NIGHT LIGHT by early Oct; Christmas SKUs by mid-Sept.

R5: International diversification — target France/Germany; assign managers to EIRE/NL whale clients.

R6: Separate Product Revenue (£10.28M) from Operational Revenue (£362K) in all dashboards.

LONG-TERM (6-12 Months)

R7: Guest-to-registered conversion — post-purchase email flows for 15-23% Unknown customer revenue.

R8: Upsell 1,022 Loyal Customers to Champions via volume-based bundle discounts.

Target: Reduce UK revenue share from 84.6% to <70% within 2 years.

Next Steps for Management

5 Strategic Areas for Future Investigation

- ❶ Customer Lifetime Value (CLV): How much is a Champion worth over their entire relationship with the company?
- ❷ Cohort Retention Analysis: Are we successfully retaining customers acquired in different months, or is churn accelerating?
- ❸ Market Basket Analysis: Which products are frequently purchased together? (Cross-selling opportunities).
- ❹ Pricing Elasticity: Can we increase prices on high-demand B2B products without losing volume?
- ❺ Channel Attribution: Where do 'Unknown' customers come from, and can we convert them more efficiently?

Thank You

Questions & Discussion

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Analysis conducted using Python, Pandas, Matplotlib & Seaborn. All findings are evidence-based and derived from the UCI Online Retail dataset (ID: 352).